

8.2 Integration by Parts

Integration by parts is a technique for simplifying integrals of the form

$$\int u(x)v'(x) dx$$

It is useful when u can be differentiated repeatedly and v' can be integrated repeatedly without difficulty. The integrals

$$\int x \cos x dx \text{ and } \int x^2 e^x dx$$

are such integrals because $u(x) = x$ or $u(x) = x^2$ can be differentiated repeatedly to become zero, and $v'(x) = \cos x$ or $v'(x) = e^x$ can be integrated repeatedly without difficulty. Integration by parts also applies to integrals like

$$\int \ln x dx \text{ and } \int e^x \cos x dx.$$

In the first case, the integrand $\ln x$ can be rewritten as $(\ln x)(1)$, and $u(x) = \ln x$ is easy to differentiate while $v'(x) = 1$ easily integrates to x . In the second case, each part of the integrand appears again after repeated differentiation or integration.

① Product Rule in Integral Form

If u and v are differentiable functions of x , the Product Rule says that

$$\frac{d}{dx}(uv) = u'v + uv'.$$

In terms of indefinite integrals, this equation becomes

$$\int \frac{d}{dx}(uv) dx = \int (u'v + uv') dx$$

or

$$uv = \int u'v dx + \int uv' dx.$$

■ Integration by Parts Formula

$$\int uv' dx = uv - \int vu' dx.$$

■ Integration by Parts Formula—Differential Version

$$\int u \, dv = uv - \int v \, du.$$

② Tip to choose u and dv in $\int f(x) \, dx$

(a) Try letting u be a factor of $f(x)$ and u' is a function simple than u .

(b) Try letting dv be the most complicated portion of $f(x)dx$ and $\int dv$ can be found by a basic integration formula.

Example 1 Find

$$\int x \cos x \, dx.$$

Example 2 Find

$$\int \ln x \, dx.$$

Example 3 Evaluate

$$\int x e^x \, dx.$$

Example 4 Evaluate

$$\int x^2 e^x \, dx.$$

Example 5 Evaluate

$$\int e^x \cos x \, dx.$$

Example 6 Evaluate

$$\int e^x \sin x \, dx.$$

Example 7 Evaluate

$$\int x^2 \ln x \, dx.$$

Example 8 Evaluate

$$\int (x^2 + 5x) e^x \, dx.$$